

ECONOMETRIC THEORY AND PRACTICE I

Quiz 4

3 questions | 10 minutes | 1 point each | No negative marking



Keep the QR code and corner marks clear.

form ref: 4H8M

Fill all 3 ID digits clearly. An unassignable ID receives zero.

STUDENT ID - 3 DIGITS

Write each digit, then fill its bubble.

	digit 1	digit 2	digit 3
0	☐	☐	☐
1	☐	☐	☐
2	☐	☐	☐
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RESPONSE PANEL

Fill exactly one bubble in each row.

Question 1	☐ A	☐ B	☐ C	☐ D	☐ E	☐ F	☐ G
Question 2	☐ A	☐ B	☐ C	☐ D	☐ E	☐ F	☐ G
Question 3	☐ A	☐ B	☐ C	☐ D	☐ E	☐ F	☐ G

1. The causal model is

$$Y = \alpha + 7X + 3Z + \varepsilon, \quad E(\varepsilon \mid X, Z) = 0.$$

Suppose $\text{Cov}(X, Z) = -2$ and $\text{Var}(X) = 6$. What is the population slope from regressing Y on X alone, with an intercept?

- | | |
|-------------------|--------|
| A. 8 | B. -1 |
| C. $\frac{20}{3}$ | D. 1 |
| E. 6 | F. 6.5 |
| G. 7 | |

2. For exogenous OLS, write

$$\sqrt{n}(\hat{b} - \beta) = \frac{A_n}{D_n}, \quad D_n = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2.$$

Suppose $\text{Var}(X) = 4$ and $A_n \xrightarrow{d} \mathcal{N}(0, 144)$, where the second normal parameter is the variance. Under the lecture's LLN/CLT conditions, which pair gives (probability limit of D_n ; approximate $\text{SD}(\hat{b})$ at $n = 100$)?

- | | |
|----------------|-------------|
| A. (2; 0.6) | B. (4; 0.3) |
| C. (4; 0.03) | D. (4; 0.6) |
| E. (16; 0.075) | F. (4; 3) |
| G. (4; 0.09) | |

3. In the causal model,

$$Y = \alpha + \beta X + 3Z + \varepsilon, \quad E(\varepsilon \mid X, Z) = 0.$$

Suppose $\text{SD}(X) = 2$, $\text{SD}(Z) = 4$, and $\text{Corr}(X, Z) = \frac{1}{2}$. The population slope of Y on X alone (with an intercept) is $b_{\text{short}} = 0.5$. Under i.i.d. sampling and the LLN conditions, which pair gives

$$\left(\text{causal } \beta; \lim_{n \rightarrow \infty} \Pr(\hat{b}_{\text{short}} > 0) \right)?$$

- | | |
|--------------|----------------|
| A. (-2.5; 1) | B. (3.5; 1) |
| C. (0.5; 1) | D. (-2.5; 0.5) |
| E. (-1; 1) | F. (-11.5; 1) |
| G. (-2.5; 0) | |

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STUDENT ID - 3 DIGITS

Write each digit, then fill its bubble.

	digit 1	digit 2	digit 3
0	☐	☐	☐
1	☐	☐	☐
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5	☐	☐	☐
6	☐	☐	☐
7	☐	☐	☐
8	☐	☐	☐
9	☐	☐	☐

RESPONSE PANEL

Fill exactly one bubble in each row.

Question 1	☐ A	☐ B	☐ C	☐ D	☐ E	☐ F	☐ G
Question 2	☐ A	☐ B	☐ C	☐ D	☐ E	☐ F	☐ G
Question 3	☐ A	☐ B	☐ C	☐ D	☐ E	☐ F	☐ G

1. The causal model is

$$Y = \alpha + 8X + 3Z + \varepsilon, \quad E(\varepsilon \mid X, Z) = 0.$$

Suppose $\text{Cov}(X, Z) = -2$ and $\text{Var}(X) = 6$. What is the population slope from regressing Y on X alone, with an intercept?

- | | |
|------|-------------------|
| A. 2 | B. -1 |
| C. 9 | D. 7.5 |
| E. 8 | F. $\frac{23}{3}$ |
| G. 7 | |

2. For exogenous OLS, write

$$\sqrt{n}(\hat{b} - \beta) = \frac{A_n}{D_n}, \quad D_n = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2.$$

Suppose $\text{Var}(X) = 4$ and $A_n \xrightarrow{d} \mathcal{N}(0, 256)$, where the second normal parameter is the variance. Under the lecture's LLN/CLT conditions, which pair gives (probability limit of D_n ; approximate $\text{SD}(\hat{b})$ at $n = 100$)?

- | | |
|--------------|--------------|
| A. (4; 4) | B. (4; 0.16) |
| C. (4; 0.8) | D. (2; 0.8) |
| E. (4; 0.04) | F. (4; 0.4) |
| G. (16; 0.1) | |

3. In the causal model,

$$Y = \alpha + \beta X + 3Z + \varepsilon, \quad E(\varepsilon \mid X, Z) = 0.$$

Suppose $\text{SD}(X) = 2$, $\text{SD}(Z) = 4$, and $\text{Corr}(X, Z) = \frac{1}{2}$. The population slope of Y on X alone (with an intercept) is $b_{\text{short}} = 1$. Under i.i.d. sampling and the LLN conditions, which pair gives

$$\left(\text{causal } \beta; \lim_{n \rightarrow \infty} \Pr(\hat{b}_{\text{short}} > 0) \right)?$$

- | | |
|--------------|-------------|
| A. (-2; 0.5) | B. (4; 1) |
| C. (-2; 1) | D. (-11; 1) |
| E. (-2; 0) | F. (1; 1) |
| G. (-0.5; 1) | |

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form ref: 423M

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	digit 1	digit 2	digit 3
0	☐	☐	☐
1	☐	☐	☐
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4	☐	☐	☐
5	☐	☐	☐
6	☐	☐	☐
7	☐	☐	☐
8	☐	☐	☐
9	☐	☐	☐

RESPONSE PANEL

Fill exactly one bubble in each row.

Question 1	☐ A	☐ B	☐ C	☐ D	☐ E	☐ F	☐ G
Question 2	☐ A	☐ B	☐ C	☐ D	☐ E	☐ F	☐ G
Question 3	☐ A	☐ B	☐ C	☐ D	☐ E	☐ F	☐ G

1. The causal model is

$$Y = \alpha + 9X + 3Z + \varepsilon, \quad E(\varepsilon \mid X, Z) = 0.$$

Suppose $\text{Cov}(X, Z) = -2$ and $\text{Var}(X) = 6$. What is the population slope from regressing Y on X alone, with an intercept?

- | | |
|-------------------|--------|
| A. $\frac{26}{3}$ | B. 8.5 |
| C. -1 | D. 3 |
| E. 9 | F. 8 |
| G. 10 | |

2. For exogenous OLS, write

$$\sqrt{n}(\hat{b} - \beta) = \frac{A_n}{D_n}, \quad D_n = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2.$$

Suppose $\text{Var}(X) = 4$ and $A_n \xrightarrow{d} \mathcal{N}(0, 400)$, where the second normal parameter is the variance. Under the lecture's LLN/CLT conditions, which pair gives (probability limit of D_n ; approximate $\text{SD}(\hat{b})$ at $n = 100$)?

- | | |
|--------------|----------------|
| A. (4; 0.25) | B. (4; 0.05) |
| C. (2; 1) | D. (4; 1) |
| E. (4; 5) | F. (16; 0.125) |
| G. (4; 0.5) | |

3. In the causal model,

$$Y = \alpha + \beta X + 3Z + \varepsilon, \quad E(\varepsilon \mid X, Z) = 0.$$

Suppose $\text{SD}(X) = 2$, $\text{SD}(Z) = 4$, and $\text{Corr}(X, Z) = \frac{1}{2}$. The population slope of Y on X alone (with an intercept) is $b_{\text{short}} = 1.5$. Under i.i.d. sampling and the LLN conditions, which pair gives

$$\left(\text{causal } \beta; \lim_{n \rightarrow \infty} \Pr(\hat{b}_{\text{short}} > 0) \right)?$$

- | | |
|---------------|----------------|
| A. (0; 1) | B. (1.5; 1) |
| C. (-10.5; 1) | D. (-1.5; 1) |
| E. (4.5; 1) | F. (-1.5; 0.5) |
| G. (-1.5; 0) | |

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2	☐	☐	☐
3	☐	☐	☐
4	☐	☐	☐
5	☐	☐	☐
6	☐	☐	☐
7	☐	☐	☐
8	☐	☐	☐
9	☐	☐	☐

RESPONSE PANEL

Fill exactly one bubble in each row.

Question 1	☐ A	☐ B	☐ C	☐ D	☐ E	☐ F	☐ G
Question 2	☐ A	☐ B	☐ C	☐ D	☐ E	☐ F	☐ G
Question 3	☐ A	☐ B	☐ C	☐ D	☐ E	☐ F	☐ G

1. The causal model is

$$Y = \alpha + 10X + 3Z + \varepsilon, \quad E(\varepsilon | X, Z) = 0.$$

Suppose $\text{Cov}(X, Z) = -2$ and $\text{Var}(X) = 6$. What is the population slope from regressing Y on X alone, with an intercept?

- | | |
|--------|-------------------|
| A. 9 | B. 4 |
| C. 9.5 | D. $\frac{29}{3}$ |
| E. -1 | F. 10 |
| G. 11 | |

2. For exogenous OLS, write

$$\sqrt{n}(\hat{b} - \beta) = \frac{A_n}{D_n}, \quad D_n = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2.$$

Suppose $\text{Var}(X) = 4$ and $A_n \xrightarrow{d} \mathcal{N}(0, 576)$, where the second normal parameter is the variance. Under the lecture's LLN/CLT conditions, which pair gives (probability limit of D_n ; approximate $\text{SD}(\hat{b})$ at $n = 100$)?

- | | |
|---------------|--------------|
| A. (4; 6) | B. (4; 1.2) |
| C. (4; 0.36) | D. (4; 0.6) |
| E. (2; 1.2) | F. (4; 0.06) |
| G. (16; 0.15) | |

3. In the causal model,

$$Y = \alpha + \beta X + 3Z + \varepsilon, \quad E(\varepsilon | X, Z) = 0.$$

Suppose $\text{SD}(X) = 2$, $\text{SD}(Z) = 4$, and $\text{Corr}(X, Z) = \frac{1}{2}$. The population slope of Y on X alone (with an intercept) is $b_{\text{short}} = 2$. Under i.i.d. sampling and the LLN conditions, which pair gives

$$\left(\text{causal } \beta; \lim_{n \rightarrow \infty} \Pr(\hat{b}_{\text{short}} > 0) \right)?$$

- | | |
|-------------|--------------|
| A. (-1; 0) | B. (-1; 1) |
| C. (0.5; 1) | D. (-1; 0.5) |
| E. (5; 1) | F. (-10; 1) |
| G. (2; 1) | |

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0	☐	☐	☐
1	☐	☐	☐
2	☐	☐	☐
3	☐	☐	☐
4	☐	☐	☐
5	☐	☐	☐
6	☐	☐	☐
7	☐	☐	☐
8	☐	☐	☐
9	☐	☐	☐

RESPONSE PANEL

Fill exactly one bubble in each row.

Question 1	☐ A	☐ B	☐ C	☐ D	☐ E	☐ F	☐ G
Question 2	☐ A	☐ B	☐ C	☐ D	☐ E	☐ F	☐ G
Question 3	☐ A	☐ B	☐ C	☐ D	☐ E	☐ F	☐ G

1. The causal model is

$$Y = \alpha + 11X + 3Z + \varepsilon, \quad E(\varepsilon | X, Z) = 0.$$

Suppose $\text{Cov}(X, Z) = -2$ and $\text{Var}(X) = 6$. What is the population slope from regressing Y on X alone, with an intercept?

- | | |
|---------|-------------------|
| A. 11 | B. $\frac{32}{3}$ |
| C. 12 | D. 10 |
| E. 10.5 | F. 5 |
| G. -1 | |

2. For exogenous OLS, write

$$\sqrt{n}(\hat{b} - \beta) = \frac{A_n}{D_n}, \quad D_n = \frac{1}{n} \sum_{i=1}^n (X_i - \bar{X})^2.$$

Suppose $\text{Var}(X) = 4$ and $A_n \xrightarrow{d} \mathcal{N}(0, 784)$, where the second normal parameter is the variance. Under the lecture's LLN/CLT conditions, which pair gives (probability limit of D_n ; approximate $\text{SD}(\hat{b})$ at $n = 100$)?

- | | |
|--------------|----------------|
| A. (4; 7) | B. (4; 0.07) |
| C. (4; 0.7) | D. (2; 1.4) |
| E. (4; 0.49) | F. (16; 0.175) |
| G. (4; 1.4) | |

3. In the causal model,

$$Y = \alpha + \beta X + 3Z + \varepsilon, \quad E(\varepsilon | X, Z) = 0.$$

Suppose $\text{SD}(X) = 2$, $\text{SD}(Z) = 4$, and $\text{Corr}(X, Z) = \frac{1}{2}$. The population slope of Y on X alone (with an intercept) is $b_{\text{short}} = 2.5$. Under i.i.d. sampling and the LLN conditions, which pair gives

$$\left(\text{causal } \beta; \lim_{n \rightarrow \infty} \Pr(\hat{b}_{\text{short}} > 0) \right)?$$

- | | |
|--------------|----------------|
| A. (-0.5; 0) | B. (-0.5; 0.5) |
| C. (1; 1) | D. (2.5; 1) |
| E. (-0.5; 1) | F. (5.5; 1) |
| G. (-9.5; 1) | |